Effects of individual quotas on catching efficiency and spawning potential in the Alaska sablefish fishery

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Abstract: Individual fishery quota (IFQ) management eliminates the race for fish and may improve economic efficiency, conservation, and safety in fisheries. Empirical information documenting these effects is limited, even though IFQs have been used since the late 1970s. We analyzed fishery data from the Alaska sablefish (*Anoplopoma fimbria*) longline fishery, which came under IFQ management in 1995. We compared the fishery data with fishery-independent survey data, which acted as a control to separate annual changes in population demographics from changes due to IFQ management. We found that IFQ management increased fishery catch rate and decreased harvest of immature fish. Catching efficiency increased 1.8 times with the change from an open-access to an IFQ fishery. The improved catching efficiency of the IFQ fishery reduced variable costs to catch the quota from 8 to 5% of landed value, a savings averaging $3.1 million US annually. Decreased harvest of immature fish improved the chance that individual fish will reproduce at least once. Spawning potential of sablefish, expressed as spawning biomass per recruit, increased 9% for the IFQ fishery. Switching from the open-access fishery's race for fish to IFQs provided two clear benefits that should be considered when evaluating management options for other open-access fisheries.

Résumé : La gestion des pêches commerciales à l'aide de quotas individuels (IFQ) élimine la compétition pour les poissons et peut probablement augmenter l'efficacité économique, la conservation et la sécurité de la pêche. Il y a peu de données empiriques sur ces effets, bien que les IFQ soient en usage depuis la fin des années 1970. Nous analysons ici les statistiques de la pêche à la palangre de la Morue charbonnière (*Anoplopoma fimbria*) en Alaska qui est gérée à l'aide d'IFQ depuis 1995. Une comparaison des données provenant de la pêche commerciale avec celles d'autres inventaires indépendants de la pêche et utilisées comme témoins a permis de séparer les effets de la gestion par IFQ des changements démographiques annuels de la population. Il ressort de ces analyses que la gestion par IFQ augmente les taux de capture de la pêche et décroît la prise de poissons immatures. L'efficacité de la capture a augmenté d'un facteur de 1,8 lorsque l'accès libre a été remplacé par les quotas individuels. Cette augmentation d'efficacité a réduit les frais variables reliés à l'atteinte des quotas de 8 à 5% de la valeur des poissons débarqués, une économie de 3,1 millions $US chaque année. La diminution de la récolte de poissons immatures augmente la probabilité que chaque poisson se reproduise au moins une fois. Le potentiel de fraye de la Morue charbonnière, défini comme la biomasse des géniteurs par recrue, a augmenté de 9%. Le remplacement d'une pêche à accès libre qui entraîne une compétition pour les poissons par une pêche gérée par IFQ produit donc deux bénéfices nets qui doivent être pris en compte lorsqu'on évalue les options de gestion dans un pêche commerciale à accès libre.

[Traduit par la Rédaction]

Introduction

Groundfish fisheries in Alaska are managed by annual catch quotas, limited access, bycatch measures including catch limits, and closed areas triggered by reaching these limits. Since 1995, the sablefish (*Anoplopoma fimbria*) and Pacific halibut (*Hippoglossus stenolepis*) longline fisheries also have been managed by individual (fishery) quotas or IFQs. An annual catch quota sets the overall level of harvest, whereas an individual quota allocates the quota among individual fishermen. The Magnuson–Stevens Act defines an IFQ as "a Federal permit under a limited access system to harvest a quantity of fish, expressed by a unit or units representing a percentage of the total allowable catch of a fishery that may be received or held for exclusive use by a person."

IFQs have been used in some countries since the late 1970s and in a few fisheries in the United States since 1983 (National Research Council (NRC) 1999; Wertheimer and Swanson 1999). Their purpose is to address problems of common property fisheries where individual ownership is not assigned, eliminating the race for fish and improving economic efficiency, conservation, and safety (NRC 1999). The North Pacific Fishery Management Council considered IFQs for sablefish and halibut longline fisheries due to allocation and gear conflicts, ghost fishing of lost gear, discard mortality, product quality, and safety (NRC 1999).

Some empirical information exists on the economic, con-

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servation, and safety effects of IFQs. These effects include nearly year-round availability of fresh rather than frozen fish, increased ex-vessel value, and reduced crew sizes (Casey et al. 1995), less lost and abandoned gear and reduced fishing mortality from lost and abandoned gear (NRC 1999), reduced quota overages (NRC 1999, table G.2), reduced discard of bycatch (NRC 1999), increased management and enforcement costs (NRC 1999), changes in fleet behavior (Sullivan and Rebert 1998), and reduced search and rescue missions (NRC 1999, table 3.3).

Our analysis focuses on catching efficiency and spawning potential in the Alaska sablefish fishery; our purpose was to examine how IFQ management affected these two aspects of the fishery rather than to conduct an overall evaluation of sablefish IFQ management. We describe two benefits of IFQ management for sablefish, an increase in fishery catch rates and a decrease in the fraction of immature fish captured, which coincided with the switch from open-access to IFQ management. We compare fishery data from before and after inception of IFQ management to fisheries-independent survey data collected through the same time period. The survey data in effect provide for a controlled experiment to separate annual changes in population demographics from changes due to IFQ management. We further analyze the fishery data to determine what changes in fishing practices lead to these benefits.

Alaska sablefish longline fishery

Sablefish have been exploited since the end of the nineteenth century by U.S. and Canadian longline fishermen, developing as a secondary activity of the halibut longline fishery of the United States and Canada. Catches were relatively small, averaging 1700 metric tons (t) from 1930 to 1957, and generally limited to areas near fishing ports (Low et al. 1976). Japanese longliners began operations in the eastern Bering Sea in 1958 and in a few years shifted to the Aleutian Islands region and Gulf of Alaska after their fishing grounds in the eastern Bering were preempted by expanding Japanese trawl fisheries. Catches peaked at over 50 000 t in 1972 (Low et al. 1976), but the heavy fishing by foreign vessels led to a substantial population decline and fishery regulations that reduced catches to about one fifth of their peak (Sigler et al. 1999). The U.S. longline fishery began expanding in 1982 in the Gulf of Alaska and in 1988 harvested all sablefish taken in Alaska except minor joint venture catches. The expansion of the U.S. fishery during the 1980s was helped by exceptional recruitment during the late 1970s (Sigler et al. 1999). This exceptional recruitment fueled an increase in abundance for the population that had been depleted by the heavy fishing during the 1970s. Increased abundance led to relaxation of fishing quotas and catches peaked again in 1988, following the first peak in 1972. Abundance and catches have since fallen, as the exceptional late 1970s year-classes have died off, and in 1998, catches were about 37% of the 1988 peak (Sigler et al. 1999).

Most (~88% from 1995 to 1998) sablefish were caught by longline (Sigler et al. 1999). Some (~12%) were caught as bycatch during directed trawl fisheries for other species groups such as rockfish and deepwater flatfish, and less than 1% were caught by pots. Only the longline fishery is part of the IFQ program. Catches in the United States Exclusive Economic Zone (U.S. EEZ), the area inside the IFQ program, averaged 25 000 t per year from 1985 to 1998 and 17 000 t per year from 1995 to 1998. Five Alaska state fisheries that are not part of the IFQ program also land sablefish, averaging 3000 t per year from 1985 to 1998.

Season length began to decrease after the change from a primarily foreign fishery to a completely domestic fishery. The year-round season in the Gulf of Alaska, the area responsible for the majority of the catches, became progressively shorter beginning in 1984 (Table 1). By the late 1980s, the race for fish caused season length to decrease to 1 or 2 months (the season was closed once the quota was reached in the open-access fishery). In some areas, this open-access fishery was as short as 9 days, warranting the label “derby” fishery. For example, from 1984 to 1987, the season length decreased from 8 months to less than 2 weeks as the number of vessels increased fivefold in the West Yakutat region (NRC 1999, fig. 1.1B). Season length continued to decrease until IFQs were implemented in 1995. Under IFQ management, the season was set at 8 months.

Predicting when to close the open-access fishery was difficult because of the speed and brevity of the fishery. Often the initial open period was followed by a “mop-up” fishery if the quota had not been reached in the initial open period. The sablefish quota frequently was exceeded in the Gulf of Alaska (Table 2). In contrast, the sablefish quota has never been exceeded during IFQ management. The halibut quota also was exceeded frequently during the open-access fishery but not during the IFQ fishery (NRC 1999, table G.2).

Sablefish biology

Sablefish inhabit the northeastern Pacific Ocean from northern Mexico to the Gulf of Alaska, westward into the Aleutian Islands, and into the Bering Sea (Wolotira et al. 1993). Adult sablefish occur along the upper continental slope, in gullies that cross the continental shelf (cross-shelf gullies), and in deep fjords, generally at depths greater than 200 m. Sablefish appear to form two populations, a northern or Alaska population and a southern or west coast population, with mixing of the two populations occurring off southwest Vancouver Island and northwest Washington (Kimura et al. 1998). Alaska sablefish are highly migratory for at least part of their life, and substantial movement has been documented (Heifetz and Fujioka 1991; Kimura et al. 1998). Sablefish are a long-lived demersal species; ages over 40 years commonly are recorded (Kimura et al. 1993). Sablefish grow rapidly in early life, reaching average maximum lengths and weights of 69 cm and 3.4 kg for males and 82 cm and 6.1 kg for females. Fifty percent of females are mature at 65 cm, while 50% of males are mature at 57 cm (Sasaki 1985), corresponding to ages of 6.5 years for females and 5 years for males.

Data sources and methods

Catch, effort, and length data were collected at sea during commercial fishing operations and during fisheries-independent longline surveys. Catch per hook was computed from the catch and effort data. The mature fraction of the catch was computed from the length data. The computed values of survey and fishery catch per hook were compared during the open-access and IFQ fisheries to determine if IFQ fishery efficiency was higher. Mature fraction values also were compared to determine if fewer immature fish

Table 1. Season length (days) by region and year, 1983–1998.

Year	Bering Sea	Aleutian Islands	Western	Central	West Yakutat	East Yakutat/Southeast
1983	365	365	365	365	365	222
1984	366	366	366	255	245	181
1985	365	247	176	143	135	111
1986	na	na	94	56	40	17
1987	266	227	69	59	15	9
1988	366	163	99	73	32	32
1989	211	36	104	57	17	17
1990	365	214	92	59	16	20
1991	365	365	55	34	27	11
1992	237	289	83	20	16	9
1993	301	365	21	20	16	10
1994	220	220	11	11	11	11
1995	319	319	319	319	319	319
1996	319	319	319	319	319	319
1997	319	319	319	319	319	319
1998	319	319	319	319	319	319

Note: The Western, Central, West Yakutat, and East Yakutat/Southeast regions are located in the Gulf of Alaska. na, not available.

Table 2. Frequency and magnitude of sablefish overharvests in the Gulf of Alaska, 1985–1998 (National Marine Fisheries Service, Alaska Regional Office, Juneau, Alaska).

Year	Frequency (%)	Average overharvest (%)
1985–1994	75	5
1995–1998	0	–9

were caught during the IFQ fishery. A detailed description of data sources and methods follows.

Fishery

From 1990 to 1998, at-sea fishery observers recorded data from 16 133 longline sets targeting sablefish, averaging 4800 hooks per set. Catch, effort, and length data were collected. Observers only collected length data from the predominant species in a set. Observers collected length data before catch sorting so discarding would not affect length samples. A set was defined as targeting sablefish where sablefish were at least 50% of the catch by weight. Fishermen also recorded catch and effort but no length data for 538 longline sets targeting sablefish through a voluntary sablefish logbook program initiated in 1997. The latter program collected data from smaller boats not required to carry an observer.

The logbook reported weights usually are approximate because vessel captains typically estimated and recorded catch for each set in the logbook while at sea and without an accurate scale measurement. An accurate weight for the entire trip was measured at landing and recorded as the IFQ landing report. We adjusted the captain's estimate of catch per set using the ratio of IFQ landing report and logbook reported weight.

The fishery catch rates were standardized to account for differences in hook spacing. Hook spacings of about 1 m (reported spacings of 39 in. and 42 in.) were the most common spacings used in the directed sablefish longline fishery, accounting for 64% of all observed sets from 1990 to 1998. We chose the 1-m spacing as the standard, a spacing that we have tested in sablefish hook spacing experiments. Two hook spacing experiments have been conducted, one from 4 to 13 July 1986 in Chatham Strait, Alaska, and the other from 25 to 26 July 1999 in the Gulf of Alaska off Icy Bay, Alaska. In the first experiment, three hook spacings were tested, 1, 2, and 4 m, and in the second experiment, a fourth spacing of 0.5 m was added. All hook spacings were tested in each set to account for the effect of set location on the resultant catch rates. The order of hook spacing within a set was varied systematically to control for interaction effects of treatments on catch rate. Circle hooks (size equivalent to 13/0) were used in both experiments; they were baited with herring in the first experiment and with squid in the second experiment. For both experiments, catch per hook increased as hook spacing increased (Table 3).

Skud and Hamley (1978) developed a method for standardizing halibut longline catch rates for hook spacing. This method continues to be applied by the International Pacific Halibut Commission (Sullivan et al. 1999). The Commission analysts used an asymptotic function, $\mathrm{CPUE}_h/\mathrm{CPUE}_s = C_\infty\,(1 - \exp(-kh))$, where CPUE is catch per hook, h is some hook spacing, s is the standard hook spacing, C_∞ is the maximum relative catch per hook, and k is a measure of how fast this maximum is reached. This equation implies that catch per hook increases with hook spacing but at a rate less than proportional to hook spacing. Parameter estimates from the sablefish hook spacing experiments are $C_\infty = 2.2$ and $k = 0.57$. The estimated asymptote of 2.2 is only slightly larger than the relative catch per hook of 4-m gear. This implies that the relative catch per hook is near maximum at this spacing. The number of standardized hooks N_{std} was computed from the number of unstandardized hooks N_{unstd} for each longline set from the logbook and observer databases, $N_{\mathrm{std}} = N_{\mathrm{unstd}}2.2(1 - \exp(-0.57h))$, where h is expressed in metres. A standardized CPUE was calculated for each set by dividing the weight of the catch by the standardized number of hooks, $\mathrm{CPUE}_{\mathrm{std}} = \mathrm{weight}/N_{\mathrm{std}}$.

The standardized catch rates by set were used to compute average catch rates by vessel for each North Pacific Fishery Management Council region: Bering Sea, Aleutian Islands, and the Gulf of Alaska regions Western, Central, West Yakutat, and East Yakutat/Southeast. Catch rate by vessel–region was computed by dividing each vessel's total regional catch by its total regional effort. The average for each region was computed from the vessel–region catch rates. These averages combined the observer and logbook data. The averages for the IFQ fishery were computed using only data collected during the summer survey period, which the 8-month long IFQ fishery overlapped. Selecting this subset of the data removes the effect of seasonality on the comparison of the

Table 3. Sablefish catch (kg) per hook for four different hook spacings.

		Hook spacing (m)			
Year	Set	0.5	1	2	4
1986	1	na	0.62	1.03	1.33
1986	2	na	1.10	1.33	1.40
1986	3	na	0.61	1.13	1.39
1986	4	na	0.92	1.34	1.83
1986	5	na	0.74	1.07	1.58
1986	6	na	0.93	1.00	1.51
1986	7	na	0.73	1.11	1.61
1986	8	na	1.00	1.44	1.72
1986	9	na	1.02	1.56	1.74
1986	10	na	0.92	1.37	2.08
1999	1	0.10	0.09	0.17	0.15
1999	2	0.30	0.43	0.50	0.55
1999	3	0.16	0.08	0.29	0.40
1999	4	0.26	0.43	0.91	1.30
1999	5	0.45	0.68	1.06	1.53
1999	6	0.49	0.74	1.35	1.48

Note: The 0.5-m hook spacing was not tested in 1986. na, not available.

IFQ fishery and survey. This approach cannot be used for comparisons of the open-access fishery and survey because the open-access fishery and survey did not overlap; the open-access fishery occurred during spring, while the survey occurred during summer. Thus, comparisons of the open-access fishery and survey are confounded with seasonality. We tested for seasonality by examining fishery data for the spring and summer from years where seasonal data were available, 1995–1998.

Hook size, which may vary in the commercial fishery, was not considered in our analysis because these data are not collected by the fishery observers. However, typical sablefish and cod hooks are similar in size, so any effect on catch rates and size selectivity should be small. In contrast, typical halibut hooks are about one and one-half (1.5) times larger than typical sablefish hooks. In an experiment comparing these two hook sizes, sablefish catch rates were 33% lower for the larger hooks and sablefish size frequencies were similar (M.F. Sigler, National Marine Fisheries Service, Alaska Fisheries Science Center, Auke Bay Laboratory, Juneau, Alaska, unpublished data). These results imply that fishermen typically will not use halibut hooks to fish for sablefish because of the catch rate reduction.

Longline surveys

Catch, effort, and length data were collected during sablefish longline surveys conducted annually since 1978 (Sasaki 1985; Sigler and Fujioka 1988; Sigler and Zenger 1989). The longline surveys sampled cross-shelf gullies and the upper continental slope. Sampling depths were about 150–1000 m. About 88 standard stations were sampled each year; about 66 of these are located in the usual fishing grounds during the IFQ fishery. The latter stations were selected for comparison with the fishery data. There were 8–16 stations per region. The same number of stations were sampled each year from 1990 to 1998. Each station consisted of 7200 hooks spaced 2 m apart along a groundline 16 km long, except in the Bering Sea, where each station consisted of 8100 hooks along a groundline 18 km long. Only longline surveys conducted from 1990 to 1998 in the Gulf of Alaska, in 1996 and 1998 in the Aleutian Islands, and in 1997 in the eastern Bering Sea were compared with the fishery data. The Bering Sea and Aleutian Islands also were sampled from 1990 to 1994, but with different longline gear. This lack of standard sampling of the Bering Sea and Aleutian Islands for all years from 1990 to 1998 limits our analyses for these regions.

The survey catch data were stratified by depth interval (100-m intervals from 100 to 400 m and 200-m intervals from 400 to 1000 m) to account for differences in catch rate among the surveyed depths, and then, catch rates were computed by set and depth strata. The fishery catch rates were not stratified by depth because the fishery depth information was not detailed enough. The survey catch rates by set–depth were adjusted to the fishery standard of 1-m hook spacing. Average catch rates by depth strata were computed for each management region and then combined across depth strata by weighting each depth–region catch rate by the amount of habitat area in that management region and depth strata. The resultant values were catch rates by management region and year. These habitat-weighted catch rates provide indices of sablefish relative abundance to compare with the fishery catch rates.

Fraction mature

The fraction of the sampled population (numbers) that was mature was computed from length data and prior estimates of maturity at length. A logistic equation was used to represent maturity at length, $1/(1 + e^{-\beta(L-L_{50})})$, where L is length, L_{50} is the length at 50% maturity, and β is the slope at L_{50}; L_{50} is 57 cm for males and 65 cm for females and $\beta = 0.4$ (values estimated from Sasaki 1985, fig. 23).

Distance between sets

The distance between sets is a measure of the crowding on the grounds during the open-access fishery. For each set, we determined the distance to the nearest set by the same vessel. We chose to compare sets by the same vessel for two reasons. First, only part of the fleet is observed and there usually are unobserved vessels fishing nearby the observed vessels. However, the between-set distance for the same vessel measures the influence of these unobserved vessels and the crowding on the grounds. Second, vessels typically record their fishing locations, so they are unlikely to repeat fishing a location, whereas they inadvertently may fish a location previously fished by another vessel that has left the area. We examined fishing locations as far back as the previous 7 days. We chose 7 days as the cutoff because fishermen report that this is the length of time that they typically let the grounds “rest” before returning to an area.

Results

Fishery catch rates

Fishery catch rates increased with the transition from an open-access to an IFQ fishery. Fishery catch rates by region and year averaged 0.24 kg per hook during the open-access fishery and 0.39 kg per hook during the IFQ fishery (during summer) (Table 4). Fishery catch rates increased even though sablefish abundance was decreasing; survey catch rates by region and year averaged 0.56 kg per hook during the open-access fishery and 0.43 kg per hook during the IFQ fishery. We paired the fishery and survey catch rates by region and year and computed the paired differences to control for the effect of abundance changes on the fishery catch rates. These paired differences showed that fishery catch rates averaged 0.31 kg per hook less than survey catch rates during the open-access fishery but only 0.04 kg per hook less than survey catch rates during the IFQ fishery. Fishery catch rates by region and year always were less than the corresponding survey catch rates during the open-access fishery but were similar during the IFQ fishery (Fig. 1). These

Table 4. Longline survey and longline fishery catch per hook, mature fraction, and fishery season length (months) by region, 1990–1998.

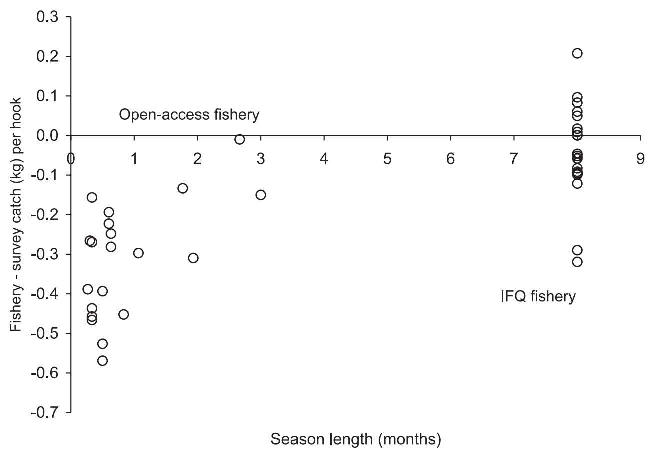
Year	Region	Season length (months)	Survey catch per hook (kg)	Fishery catch per hook (kg)	Survey male fraction mature	Fishery male fraction mature	Survey female fraction mature	Fishery female fraction mature
Open-access fishery								
1990	WGOA	3.0	0.40	0.25	0.80	0.73	0.77	0.52
1990	CGOA	1.9	0.58	0.27	0.87	0.75	0.82	0.60
1990	WYAK	0.5	0.69	0.30	0.84	0.81	0.83	0.68
1990	EYSE	0.6	0.55	0.27	0.81	na	0.80	na
1991	WGOA	1.8	0.33	0.20	0.77	0.81	0.67	0.76
1991	CGOA	1.1	0.54	0.24	0.84	0.71	0.80	0.62
1991	WYAK	0.8	0.74	0.29	0.86	0.75	0.87	0.66
1991	EYSE	0.3	0.75	0.31	0.87	0.84	0.85	0.43
1992	WGOA	2.7	0.15	0.14	0.62	0.85	0.46	0.76
1992	CGOA	0.6	0.48	0.26	0.85	0.80	0.80	0.65
1992	WYAK	0.5	0.81	0.24	0.86	0.85	0.84	0.76
1992	EYSE	0.3	0.67	0.28	0.83	0.78	0.84	0.55
1993	WGOA	0.6	0.38	0.13	0.79	0.56	0.66	0.43
1993	CGOA	0.6	0.54	0.34	0.85	0.78	0.80	0.64
1993	WYAK	0.5	0.78	0.26	0.83	0.81	0.85	0.74
1993	EYSE	0.3	0.63	0.36	0.77	0.79	0.77	0.73
1994	WGOA	0.3	0.29	0.13	0.66	0.60	0.54	0.46
1994	CGOA	0.3	0.51	0.24	0.87	0.78	0.87	0.59
1994	WYAK	0.3	0.71	0.25	0.90	0.85	0.91	0.76
1994	EYSE	0.3	0.61	0.14	0.87	0.95	0.86	0.94
Average			0.56	0.24	0.82	0.78	0.78	0.65
Average paired difference				–0.31		–0.04		–0.13
Average paired ratio				0.47				
IFQ fishery								
1995	WGOA	8.0	0.32	0.32	0.73	0.88	0.61	0.74
1995	CGOA	8.0	0.50	0.38	0.87	0.95	0.79	0.91
1995	WYAK	8.0	0.63	0.55	0.90	0.94	0.88	0.94
1995	EYSE	8.0	0.55	0.75	0.84	na	0.83	na
1996	AI	8.0	0.06	0.11	0.84	0.91	0.64	0.76
1996	WGOA	8.0	0.35	0.25	0.81	0.88	0.69	0.74
1996	CGOA	8.0	0.66	0.37	0.93	0.93	0.90	0.82
1996	WYAK	8.0	0.64	0.32	0.94	0.98	0.94	0.90
1996	EYSE	8.0	0.55	0.50	0.88	0.95	0.87	0.92
1997	EBS	8.0	0.09	0.19	0.94	na	0.80	na
1997	WGOA	8.0	0.30	0.20	0.61	0.80	0.50	0.74
1997	CGOA	8.0	0.57	0.53	0.92	0.89	0.86	0.79
1997	WYAK	8.0	0.56	0.56	0.90	0.90	0.87	0.90
1997	EYSE	8.0	0.50	0.50	0.87	0.70	0.86	0.57
1998	AI	8.0	0.09	0.15	0.76	0.86	0.57	0.71
1998	WGOA	8.0	0.33	0.23	0.72	0.91	0.63	0.74
1998	CGOA	8.0	0.44	0.38	0.91	0.90	0.88	0.86
1998	WYAK	8.0	0.46	0.54	0.90	0.87	0.90	0.88
1998	EYSE	8.0	0.49	0.51	0.88	0.94	0.81	0.97
Average			0.43	0.39	0.85	0.89	0.78	0.82
Average paired difference				–0.04		0.05		0.04
Average paired ratio				1.04				

Note: AI, Aleutian Islands; EBS, eastern Bering Sea; WGOA, Western Gulf of Alaska; CGOA, Central Gulf of Alaska; WYAK, West Yakutat; EYSE, East Yakutat/Southeast. na, not available.

paired comparisons differ significantly between the open-access and IFQ fisheries (two-sample t test, df = 37, $p < 0.001$).

The increase in fishery catch rates relative to standard survey catch rates implies that the catching efficiency of the longline fleet improved with the transition from the open-access to the IFQ fishery. A measure of the catching efficiency is the ratio of the fishery and survey catch rates. We paired the fishery and survey catch rates by region and year and divided the fishery by the survey catch rate. These paired ratios showed that fishery catch rates averaged 47% of survey catch rates during the open-access fishery and

Fig. 1. Comparisons of longline fishery and longline survey catch per hook by region and year (1990–1998) versus season length. The regions are the North Pacific Fishery Management Council regions Bering Sea, Aleutian Islands, and the Gulf of Alaska regions Western, Central, West Yakutat, and East Yakutat/Southeast.



104% of the survey catch rates during the IFQ fishery. The change from 47% during the open-access fishery to 104% during the IFQ fishery implies that catching efficiency of the longline fleet increased 2.2 times with the transition from an open-access to an IFQ fishery.

Seasonal variation in catch rates probably inflated the estimate of how much catching efficiency improved. The open-access fishery and survey did not overlap; the fishery occurred during spring and the survey occurred during summer. We tested for seasonal variation in catch rates by examining IFQ fishery data for spring and summer from 1995 to 1998, years where seasonal data were available. We paired the spring and summer fishery catch rates by region and year and computed the paired differences. These paired differences showed that spring catch rates averaged 0.16 kg per hook less than summer catch rates, a significant difference (paired t test, df = 23, $p = 0.001$). A measure of the improved catching efficiency due to season is the ratio of the spring and summer catch rates. We paired the spring and summer catch rates by region and year and divided the spring by the summer catch rate. These paired ratios showed that spring catch rates averaged 82% of summer catch rates. Adjusting catching efficiency for the seasonal effect, estimated open-access fishery catch rates during summer averaged 57% of survey catch rates rather than 47%. Instead of increasing 2.2 times, the estimate of improved catching efficiency with the transition from an open-access to an IFQ fishery was 1.81 times.

The estimate of improved catching efficiency applies to the Gulf of Alaska but not to the Bering Sea and Aleutian Islands. The survey longline gear changed after 1994 for these two regions, so there are no control data from 1990 to 1998 to compare with the fishery data and estimate catching efficiency. Although no estimate is available, there is some evidence that catching efficiency did not change in the Bering Sea and Aleutian Islands. From 1994 to 1995, average fishery catch rate increased 111% in the Gulf of Alaska but only 9% in the Bering Sea and Aleutian Islands. The lack of a large increase for the Bering Sea and Aleutian Islands implies that catching efficiency remained the same, so for the rest of the paper, we assume that catching efficiency increased in the Gulf of Alaska but remained the same in the Bering Sea and Aleutian Islands.

Improved catching efficiency during IFQ management reduced the number of hooks and days at sea necessary to catch the quota. For example, in 1995, vessels deployed 50 million hooks during 4800 vessel-days (Table 5). Applying the relative efficiency of the IFQ and open-access fisheries, an open-access fishery would have required 80 million hooks and 7800 vessel-fishing days. Thus, the IFQ fishery required 3000 fewer days at sea to catch the quota. Fewer days at sea reduces variable costs such as fuel, bait, and gear. Cost data for Alaska groundfish fisheries are not available (Hiatt and Terry 1999); instead, we assumed that commercial fishery costs were the same as our longline survey costs. Fuel cost is based on 909 L per day and $\$0.264 \cdot L^{-1}$, bait cost is based on 5 kg of bait per 100 hooks and $\$0.88 \cdot kg^{-1}$, and gear cost is based on 39 000 hooks ($137 per thousand), 24 000 gangions ($95 per thousand for materials and labor), 11 500 beckets ($78 per thousand for materials and labor), and 40 boxes of groundline ($144 per box) required to replace worn gear during the annual longline survey, which deploys 546 400 hooks per year. The two longline vessel captains who reviewed our manuscript said that these values were reasonable estimates of fuel, bait, and gear costs for the commercial fishery. We estimated that commercial fishery costs for fuel, bait, and gear were $6 million for the 1995 IFQ fishery. An open-access fishery would have required $9.7 million. The landed value of sablefish in 1995 was $111 million, so IFQ management reduced these costs from 9 to 5% of landed value (Table 5). We also estimated the cost effect for the 1995–1998 fisheries. IFQ management reduced the total number of hooks fished by 38%, from 263 million to 164 million, or about 25 million hooks per year. Fuel, gear, and bait costs decreased from $33 million to $20 million, or about $3.1 million per year. Compared with landed value, these fuel, gear, and bait costs decreased from 8 to 5% of landed value.

Mature fraction

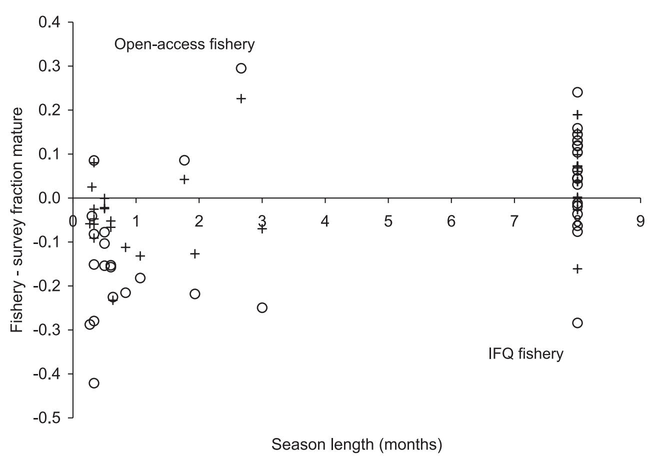
Fewer immature fish were captured with the transition from an open-access to an IFQ fishery. The difference was more obvious for females than for males. For females, fishery mature fraction by region and year averaged 0.65 during the open-access fishery and 0.82 during the IFQ fishery (during summer), an increase of 17 percentage points (Table 4). For males, fishery mature fraction by region and year averaged 0.78 during the open-access fishery and 0.89 during the IFQ fishery (during summer), an increase of 11 percentage points. Fishery mature fraction increased even though survey mature fraction increased only from 0.82 to 0.85 for males and was unchanged at 0.78 for females. We paired the fishery and survey mature fractions by sex, region, and year and computed the paired differences to control for the effect of demographic changes on the fishery mature fraction (Fig. 2). These paired differences showed that for females, fishery mature fraction averaged 13 percentage points less than survey mature fraction during the open-access fishery but was four percentage points more than survey mature fraction during the IFQ fishery, an improvement of 17 percentage points.

Table 5. Hooks, vessel-days, fuel, gear, and bait costs, their percentage of landed value, and landed value (Hiatt and Terry 1999) by year.

Year	No. of hooks deployed	No. of vessel-days	Fuel, gear, bait costs ($US)	% of landed value	Landed value ($US)
Open-access fishery					
1990	96 892 928	8844	11 424 585	13	86 000 000
1991	77 977 461	6991	9 133 668	9	97 900 000
1992	76 155 421	6931	8 969 807	10	91 900 000
1993	86 702 379	6013	9 310 659	10	97 400 000
1994	81 543 546	7170	9 483 863	8	114 400 000
Average				10	
IFQ fishery					
1995	49 681 040	4833	6 000 867	5	111 000 000
1996	44 691 335	4963	5 693 841	6	97 100 000
1997	34 356 229	3665	4 304 694	5	89 100 000
1998	35 025 506	3477	4 264 013	5	90 058 065
Average				5	
Assuming the catching efficiency of the open-access fishery					
1995	80 115 800	7793	9 677 017	9	
1996	72 395 174	8040	9 223 412	9	
1997	54 565 706	5820	6 836 858	8	
1998	56 231 048	5581	6 845 580	8	
Average				8	

Note: These values also are estimated for 1995–1998 assuming the reduced catching efficiency of the open-access fishery in the Gulf of Alaska.

Fig. 2. Comparisons of longline fishery and longline survey mature fraction by region and year (1990–1998) versus season length (months) for males (+) and females (○). The regions are the North Pacific Fishery Management Council regions Bering Sea, Aleutian Islands, and the Gulf of Alaska regions Western, Central, West Yakutat, and East Yakutat/Southeast.



For males, fishery mature fraction averaged four percentage points less than survey mature fraction during the open-access fishery but was five percentage points more than survey mature fraction during the IFQ fishery, an improvement of nine percentage points. These paired comparisons differ significantly between the open-access and IFQ fisheries for females (two-sample t test, df = 34, $p < 0.001$) and males (two-sample t test, df = 34, $p = 0.006$). The increase in fishery mature fraction relative to survey mature fraction implies that fishery selectivity of the longline fleet favored larger, older fish with the transition from the open-access to the IFQ fishery. The difference in mature fraction between the IFQ and open-access fisheries was not due to increased discards of small fish in the IFQ fishery because length sampling occurred before catch sorting and discarding.

Seasonal variation in mature fraction probably inflated the estimate of how much mature fraction increased with the transition from the open-access to the IFQ fishery. We tested for seasonal variation in mature fraction by examining fishery data for the spring and summer from years where seasonal data were available, 1995–1998. We paired the spring and summer fishery mature fractions by sex, region, and year and computed the paired differences. These paired differences showed that for both females and males, spring mature fraction averaged eight percentage points less than summer mature fraction. These paired comparisons differ significantly for males (paired t test, df = 17, $p = 0.003$) but not for females (paired t test, df = 18, $p = 0.086$). The increase in fishery mature fraction implies that larger, older fish are more available to the fishery in summer than in spring. We adjusted the mature fraction from the open-access fishery for the season effect by subtracting eight percentage points from the differences in mature fraction between the survey and open-access fishery. This adjustment reduces the improvement in mature fraction to nine percentage points for females and one percentage point for males.

The decreased harvest of immature fish in the IFQ fishery improves the chance that individual fish will reproduce at least once. We completed a spawning biomass per recruit analysis (Gabriel et al. 1989) to evaluate the effect of the reduced harvest of immature fish on the spawning potential of sablefish. A spawning biomass per recruit analysis evaluates the total contribution of a cohort to the spawning stock biomass over the cohort's lifetime, scaled by the original number of recruits, to provide a relative measure of spawning potential regardless of the absolute number of recruits. In our spawning biomass per recruit analysis, we assumed that the reduced harvest of immature fish in the IFQ fishery was due to differences in selectivity between the open-access and IFQ fisheries and that the growth, maturity at age, natural mortality, average recruitment, and average catch were the same. Selectivity refers to both selectivity of the longline gear and availability to the longline fishery. The age of 50% selection (the average age of first capture) was adjusted to match mature fraction estimated for the open-access and IFQ fisheries during summer. Values assumed for growth, maturity at age, natural mortality, slope of the selectivity function, average recruitment (13.7 million, the average value for the 1982–1995 year-classes), and average catch (17 000 t, the average value for 1995–1998) were obtained from age-structured modeling completed for the sablefish stock assessment (Sigler et al. 1999). The spawning biomass per recruit analysis showed that spawning potential was higher for the IFQ fishery compared with the open-access fishery. Spawning biomass per recruit was 9.2 kg per recruit for the open-access fishery and 10.0 kg per recruit for the IFQ fishery, a 9% increase.

Areas and depths fished

More areas were fished during the open-access fishery compared with the IFQ fishery. The upper continental slope was fished in all years regardless of fishery type, but during the open-access fishery, two major cross-shelf gullies additionally were fished (Fig. 3). Observed sets often were located in Shelikof Trough during the open-access fishery but rarely during the IFQ fishery. Observed sets often were located in Amatuli (Seward) Gully during the open-access fishery and much less commonly during the IFQ fishery, except for 1995 in the eastern half of Amatuli Gully. Fishing only the upper continental slope during the IFQ fishery implies that these grounds are preferred over the cross-shelf gullies and that fishermen used the cross-shelf gullies during the open-access fishery due to crowding on the preferred upper continental slope. Some fishermen have confirmed this change to us, reporting that they fished in the cross-shelf gullies during the open-access fishery because the preferred upper continental slope was crowded.

Most areas of the upper continental slope have been fished during both the open-access and IFQ fisheries (Fig. 3). Less fishing effort was observed in the eastern Gulf of Alaska during the open-access fishery due to lower observer coverage rather than less fishing in this area. Although some fishing effort occurs in most areas of the upper continental slope, the proportion of fishing effort near fishing ports increased during the IFQ fishery (Fig. 4).

The depths fished did not differ substantially between the open-access and IFQ fisheries. Most fish were caught between 400 and 800 m depth regardless of year, except in 1992 and 1998, when many fish were caught at about 100 m (Fig. 5).

Crowding on the fishing grounds

The distance between sets by a vessel is a measure of the crowding on the fishing grounds. We examined the distance to the nearest set during the same day and during the previous week. The frequency distributions of the minimum distances for sets during the same day were similar for the open-access and IFQ fisheries (Fig. 6). Most sets were 1–6 km apart. The highest proportion was 2 km apart, which implies that 2 km is the shortest preferred distance between sets. Sets during the previous week usually were closer during the open-access fishery. During the open-access fishery, an average of 47% of sets had another set by the same vessel within 2 km but only 31% during the IFQ fishery (two-sample t test, df = 41, $p < 0.001$) (Fig. 7).

Setting gear at or near the same spot not only reduces subsequent catch rate but also fish size. For sets by the same vessel within 2 km of another, the catch rate on the next day's set averaged 17% less (paired t test, df = 991, $p < 0.001$). The mean length on the next day's set averaged 1% less (paired t test, df = 303, $p = 0.02$).

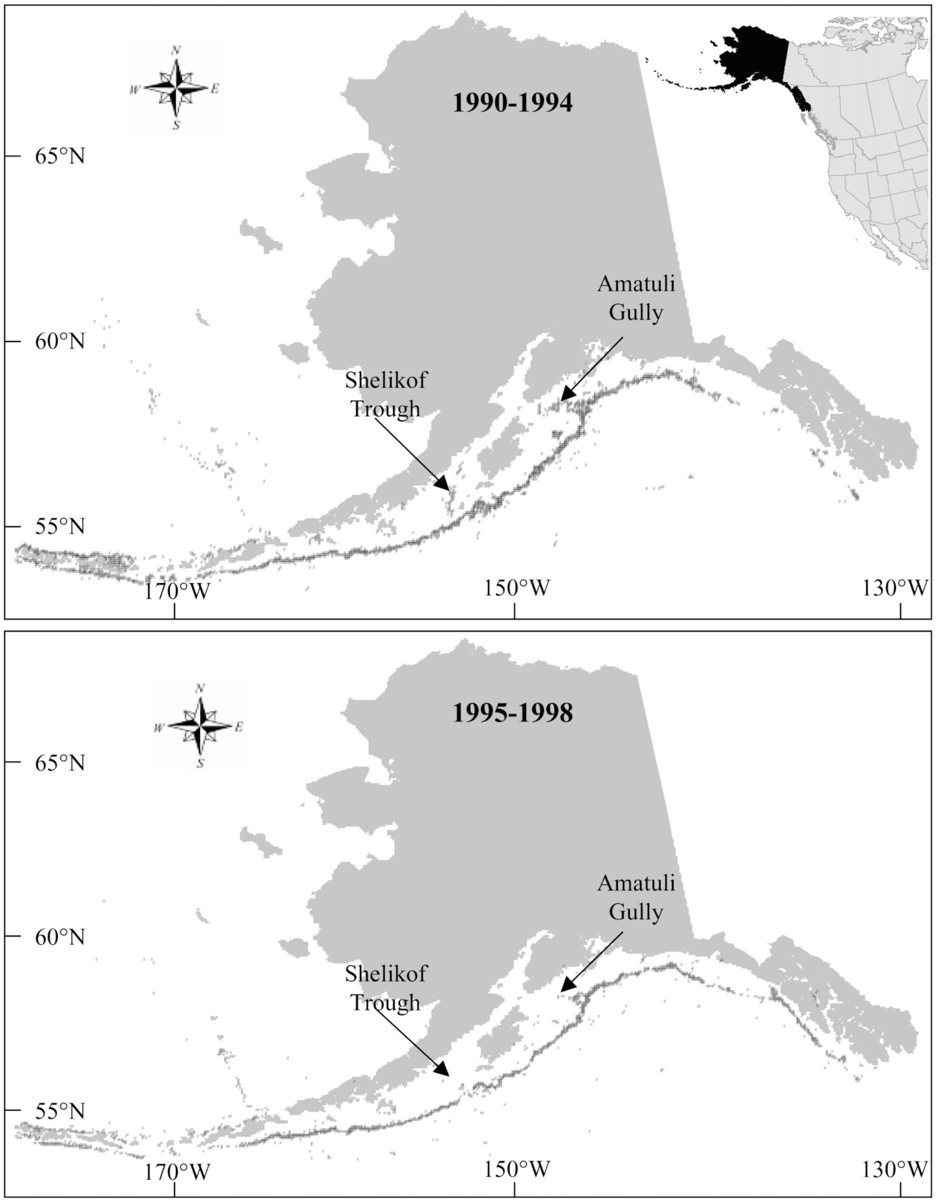
Discussion

Benefits of IFQ management

Eliminating the race for fish by switching to IFQ management increased fishery catch rates and decreased harvest of immature fish in the Alaska sablefish longline fishery. Catching efficiency increased 1.8 times, reducing variable costs to catch the quota from 8 to 5% of landed value, and saving an average of $3.1 million annually. Decreased immature harvest for the IFQ fishery improved the chance that individual fish will spawn at least once. Spawning potential of sablefish, expressed as spawning biomass per recruit, increased 9%. Improved catching efficiency is an economic benefit by decreasing variable costs associated with fishing such as fuel and bait. Improved spawning potential is a biological conservation benefit by increasing the number of fish that have a chance to spawn at least once. Catch rate estimates from the halibut fishery in British Columbia also have increased with implementation of an IFQ program, although the amount of increase is uncertain because only two scientific surveys were available for comparison and high variation and bias in their estimates limited their significance (Sullivan and Rebert 1998). These benefits support fishermen's testimony that "IFQs provided the opportunity to utilize better fishing...methods, reducing bycatch of non-targeted species..." (NRC 1999). North Pacific Fishery Management Council analysts also anticipated "improved efficiency of harvesting operations" in the Alaska sablefish and halibut fisheries once IFQ management was implemented (Pautzke and Oliver 1997).

These benefits arose because the most productive habitat was less crowded during the IFQ fishery and so the fleet no longer had to fish less productive habitat. For example, the cross-shelf gullies of Shelikof Trough and Amatuli Gully often were fished during the open-access fishery but have been fished much less often during the IFQ fishery. Sablefish are less abundant in cross-shelf gullies. Survey catch rate for Shelikof Trough averaged 61% of that for the adjacent slope from 1990 to 1994 and 52% from 1995 to 1998. The survey catch rate for Amatuli Gully averaged 80% of that for the adjacent slope from 1995 to 1998; 1995 was the first year that these gullies were fully surveyed. Sablefish also are younger in cross-shelf gullies. In 1996, the earliest year for which age information is available from the cross-shelf gullies, average age was 5 years in Shelikof Trough, about half

Fig. 3. Map of Alaska showing longline fishery set locations during the open-access fishery (1990–1994) and the IFQ fishery (1995–1998).



the age for the adjacent slope, and was 10 years in Amatuli Gully, about two thirds the age for the adjacent slope.

Less crowding on the fishing grounds with longer seasons also contributed to higher catch rates and larger fish landed. Sets on successive days by the same vessel generally were spaced closer together during the open-access fishery. By setting their gear at or near the same spot, the catch rate likely decreased due to local depletion. We found that for sets nearby one another (within 2 km), the catch rate on the next day's set averaged 17% less. Subsequent mean length was 1% less. Catch rate also decreased as the short seasons progressed in the open-access halibut fishery (Quinn 1987, table II). The rate of decrease was related to season length. The decrease ranged from 3% per day for a 20-day season to 15% per day for a 6-day season.

The sablefish fishing grounds were crowded during the open-access fishery because hundreds of vessels were fishing the limited area preferred by the fishery. The sablefish fishing grounds lie along the narrow upper continental slope of the Gulf of Alaska, Aleutian Islands, and eastern Bering Sea. The area fished during the 8-month-long IFQ fishery and presumed preferred by the fishery is about 43 000 km^2 in the Gulf of Alaska. In 1994, the last year of the open-access fishery, 1075 vessels landed sablefish in the Gulf of Alaska (Hiatt and Terry 1999), leaving about 40 km^2 per vessel. All vessels likely fished at the same time because the open-access season started on the same day, which implies a fishing area only 8 km (average set length in the directed fishery for sablefish in 1994) by 5 km for each vessel to fish during the open-access fishery.

Vessels naturally will crowd over good fishing grounds and near ports to reduce the costs of fishing. However, crowding on the grounds may prevent vessels from fishing these preferred areas. Crowding during the sablefish open-access fishery led to effort in less productive areas and farther from ports, but effort shifted to more productive areas

Fig. 4. Proportion of effort by latitude for the East Yakutat/ Southeast region for the open-access fishery (1990–1994) (◆) and the IFQ fishery (1995–1998) (+). An important fishing port, Sitka, Alaska, is located at 57°N. Latitude is rounded to the nearest 0.5°.

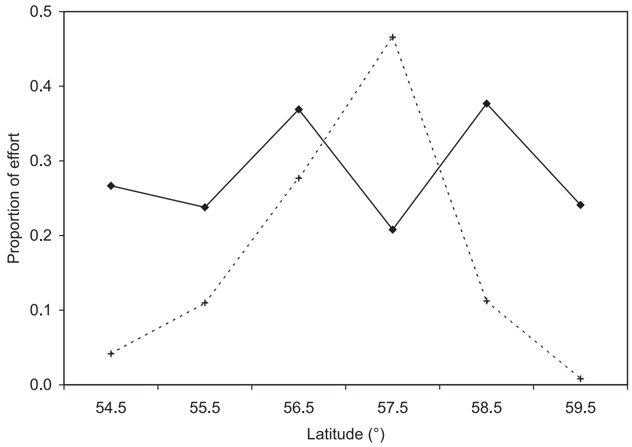


Fig. 5. Observed sablefish fishery catch versus depth by year for the open-access fishery (1990–1994) (◆) and the IFQ fishery (1995–1998) (+). Depth is rounded to the nearest 100 m.

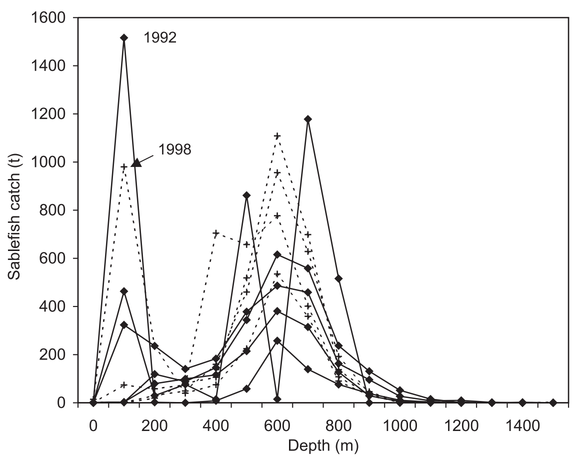


Fig. 6. Frequency distributions of the distance to the nearest observed set on the same day by the same vessel for the open-access fishery (1990–1994) (◆) and the IFQ fishery (1995–1998) (+). Distance is rounded to the nearest kilometre.

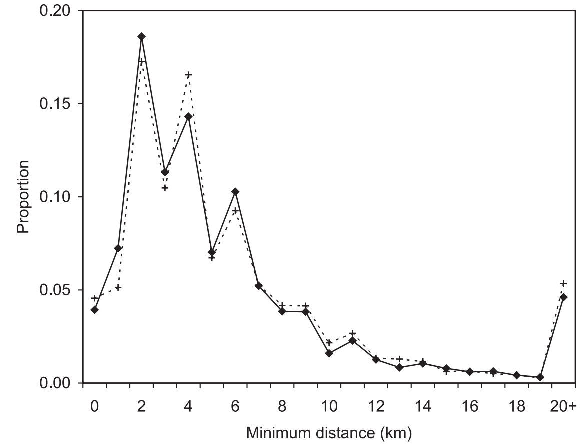
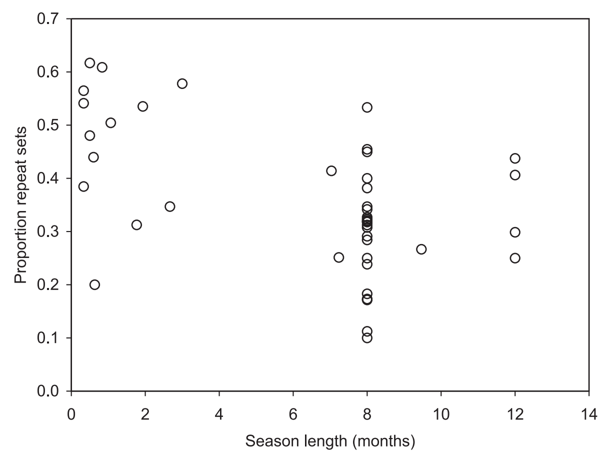


Fig. 7. For longline fishery sets by the same vessel and during the previous week, the proportion of sets by region and year where the minimum distance to the nearest set was less than or equal to 2 km (“repeat” set) versus season length. The regions are the North Pacific Fishery Management Council regions Bering Sea, Aleutian Islands, and the Gulf of Alaska regions Western, Central, West Yakutat, and East Yakutat/Southeast.



and near ports during the longer IFQ seasons. Fleet effort in the British Columbia halibut fishery also shifted to areas of higher catch rate and closer to port with implementation of an IFQ program (Sullivan and Rebert 1998), although fishermen earlier reported little change in travel distance (Casey et al. 1995).

We expected a wider range of depths to have been fished during the open-access fishery compared with the IFQ fishery, anticipating that crowding on the fishing grounds would cause some fishing in less preferred depths. However, the depth range was not larger. The lack of a discernable difference may be due to the coarseness of the position and depth data. Position and depth were recorded at only one point, the start of gear retrieval, even though the gear typically stretches over several kilometres, and may not accurately represent the distribution of depths fished. We also expected, but did not find, that the distance between sets on the same day for the same vessel would be smaller during the open-access fishery compared with the IFQ fishery. However, the between-set distances calculated from the single recorded points may not accurately represent the distribution of between-set distances. Position also was recorded only to the nearest minute of latitude and longitude, corresponding to an accuracy of about 2 km north–south and 1 km east–west at the latitude of the Alaska sablefish fishery. The position data may be too coarse to discern small differences in between-set distances. In addition, there likely is an optimum between-set distance, regardless of crowding, that minimizes travel time between sets and places sets far enough apart so as not to compete with one another.

Trip limits and fishery cooperatives

The benefits of the IFQ fishery probably accrued because of the lengthened season, which slowed the pace of the fishery. The catch occurs over several months and the vessels no longer are forced to fish side by side. An alternative management method that also lengthens the season is trip limits. A trip limit is a catch ceiling for an individual fishing trip. Unlike IFQs, trip limits do not require setting annual catch quotas. A trip limit can achieve some of the objectives possible with IFQs but also can have major detrimental impacts. A trip limit extends the supply of fish to processing plants and the consumer through a greater part of the year like an IFQ fishery. However, discarding of fish occurs when catches exceed the trip limit (Pikitch et al. 1988; Richards 1994), species may be misreported (Richards 1994), and discarding and misreporting lead to underestimates of both catch and catch rate (Richards 1994). Sablefish discards were larger for trip limit management off Oregon, 15–20% (Pikitch et al. 1988), than for IFQ management off Alaska, 2–5% (Hiatt and Terry 1999). Discarding of sablefish also may occur in the IFQ fishery when the IFQ is reached but occurs once per year rather than for every trip. Vessels averaged four IFQ sablefish trips per year during 1997 and 1998. Size-selective discarding may occur for each trip and is discussed in the "Selective fishing practices" section.

Fishery cooperatives that divide the quota into quasi-IFQs now operate in the west coast whiting and Bering Sea pollock fisheries, beginning in 1997 and 1999, respectively. Benefits associated with eliminating the race for fish also may occur for fishery cooperatives. Anticipated benefits of the whiting cooperative include more efficient allocation of resources by fishing companies, improvements in processing efficiency and product quality, and a reduction in waste and bycatch rates relative to the former "derby" fishery in which all vessels competed for a fleet-wide quota (Dorn et al. 1999). These cooperatives reportedly have slowed the pace of the fisheries, reduced overcapitalization (fewer boats chasing the fish in the pollock fishery (Anonymous 2000)), increased product recovery, and, in the case of the Bering Sea pollock fishery, made it easier for the fleet to respond to the required and prudent alternatives for protection of the Steller sea lion, which is endangered in western Alaska (North Pacific Fishery Management Council 1999).

Estimated value of increased catching efficiency

IFQ management improved catching efficiency compared with open-access management, thereby reducing the number of hooks and days at sea necessary to catch the fish. IFQ management reduced fuel, gear, and bait costs from 8 to 5% of landed value, or about $3.1 million per year, compared with open-access management. The reduction in number of hooks was 38% or about 25 million hooks per year. The race for fish during the open-access fishery clearly decreased overall benefits from the fishery. Arnason (1996) cited the five- to sixfold increase in value of the quota shares in the Icelandic demersal fisheries as evidence for improved efficiency. He hypothesized that the increased value was due to reduced capitalization and fishing effort. Our study of the sablefish IFQ fishery supports his hypothesis. We found that the slower pace of the IFQ fishery led to increased fishery efficiency and reduced fishing effort.

At the time of the North Pacific Fishery Management Council deliberations on the sablefish and halibut IFQ program, analysts estimated that the total annual benefits of the IFQ program for both sablefish and halibut would range from $30.1 to $67.6 million (Pautzke and Oliver 1997). Among these anticipated benefits were reductions in harvesting costs by $12.4 million to $13.6 million because of increased flexibility in selecting fishing strategies and the redistribution of catch and effort to lower the cost of fishing operations. The $3.1 million reduction in annual costs that we estimated for the sablefish fishery shows that at least one of these anticipated benefits of IFQs, to "improve efficiency of harvesting operations," has been achieved.

Selective fishing practices

Reducing the number of hooks required to land the catch likely also reduced habitat effects of fishing longline gear. Invertebrates such as coral, anemones, and sea stars commonly are snagged and brought to the surface with longline gear; however, documentation of how longline affects the seafloor and associated species is limited and these effects have not been quantified. High (1998) and K.J. Krieger (National Marine Fisheries Service, Alaska Fisheries Science Center, Auke Bay Laboratory, Juneau, Alaska, unpublished data) incidentally reported underwater observations of longline. High (1998) observed coral snagged by longline; he noted that unless lifted off bottom, sturdy flexible corals appeared to be relatively unharmed by the longline, whereas more fragile hard corals often had portions broken off. Krieger observed small *Primnoa* colonies attached to <0.4-m-diameter boulders that had been tipped and dragged, which he attributed to longline gear. Although the effects of longline gear are poorly known, to whatever unknown extent the habitat is affected, the reduced number of hooks deployed during the IFQ fishery reduces these habitat effects. This serendipitous effect fulfills a requirement of the Sustainable Fisheries Act of 1996, an amendment of the Magnuson–Stevens Fishery Conservation and Management Act, for fishery management councils to employ conservation and enhancement measures that minimize impacts and protect and restore that habitat.

The IFQ fishery is preferable to the open-access fishery in terms of spawning biomass per recruit; spawning potential is 9% greater for the IFQ fishery compared with the open-access fishery. Fewer immature fish were captured with the transition from an open-access to an IFQ fishery. The decreased harvest of immature fish improves the chance that individual fish will reproduce at least once.

The increased mature fraction of the catch during the IFQ fishery is not due to discards because the observer collects length information before any size sorting of the catch. Although discards did not affect our analysis, discards may occur in the IFQ fishery. During the open-access fishery, there was no motivation to discard small fish because of the short duration of the fishery and the subsequent desire to maximize catch and value. During the IFQ fishery, there is motivation to discard small fish because the value usually increases with fish size and each quota holder's catch is fixed. However, a harvester has to catch more fish to make up for highgrading, generating more operating expenses and foregoing the revenue associated with the small fish; for the sablefish IFQ fishery, highgrading is unlikely to be profitable (NRC 1999). Fisheries in which highgrading is not a serious concern seem to be characterized by minimal price differentials among fish sizes (Squires et al. 1995) and (or) relatively high costs of catching replacement fish (NRC 1999). The price differential for small sablefish is minimal, no more than 12.5% in 2000 (K. Hogan, The Auction Block, Homer, Alaska). A conservation concern also may motivate some fishermen to release small sablefish, which usually survive hooking if carefully released. There is a minimum size limit for halibut but not for sablefish. It may seem inconsistent to some fishermen to return viable, small halibut to the sea but not viable, small sablefish.

IFQ implementation

Implementing IFQs is controversial mainly because of socioeconomic concerns: the fairness of the initial allocations, the fairness of essentially privatizing harvest rights to a public resource, and the potential for consolidation of the fishery. In the United States, these and other issues have led to a moratorium on implementing new IFQ programs. On the other hand, IFQs eliminate the race for fish and the associated waste and danger of the short seasons. Our purpose was to examine how IFQ management affected catching efficiency and spawning potential in the Alaska sablefish fishery rather than to conduct an overall evaluation of sablefish IFQ management. We found that switching to IFQ management provided two clear benefits, increased fishery catch rates and decreased harvest of immature fish. These improvements occurred because less fishing takes place in marginal areas and localized depletion due to crowding of the fishing grounds is reduced. These two benefits should be weighed when considering other fisheries for IFQ management.

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